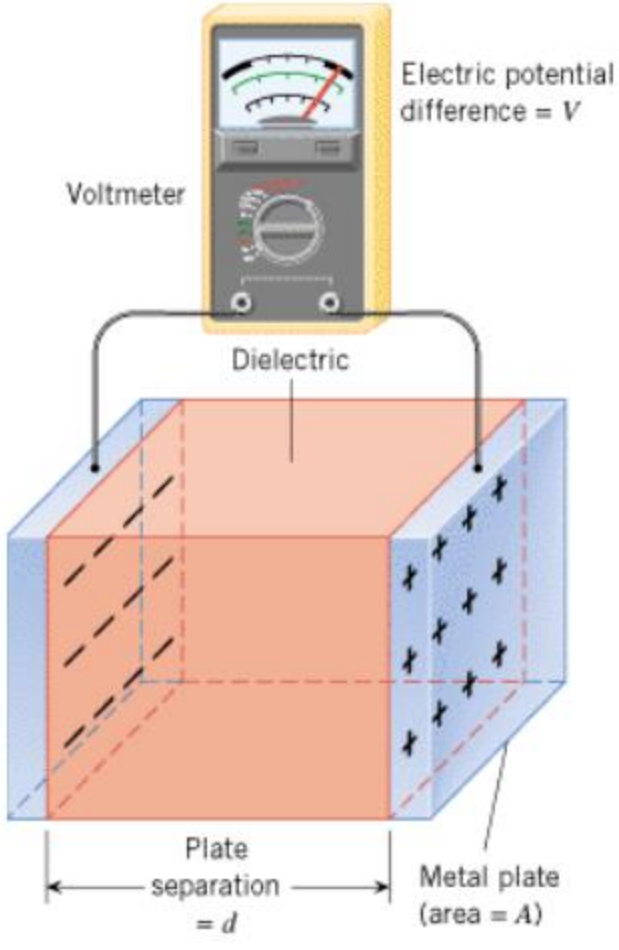
Chapter 25. Capacitance

25.1. What is Physics?
25.2. Capacitance
25.3. Calculating the Capacitance
25.4. Capacitors in Parallel and in Series
25.5. Energy Stored in an Electric Field
25.6. Capacitor with a Dielectric
25.7. Dielectrics: An Atomic View
25.8. Dielectrics and Gauss' Law

What is Physics?



- A ***capacitor*** is electric element to store electric charge .

- It consists of two conductors of any shape placed near one another without touching.

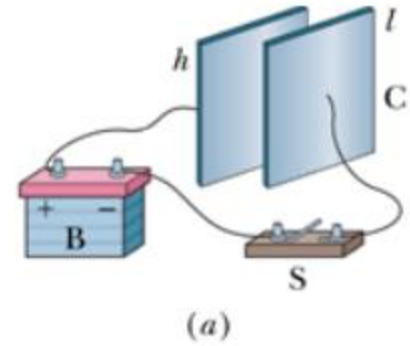
Capacitan



The magnitude q of the charge on each plate of a capacitor is directly proportional to the magnitude V of the potential difference between the plates:

$$q = CV \quad .$$

$$C = \frac{q}{V}$$

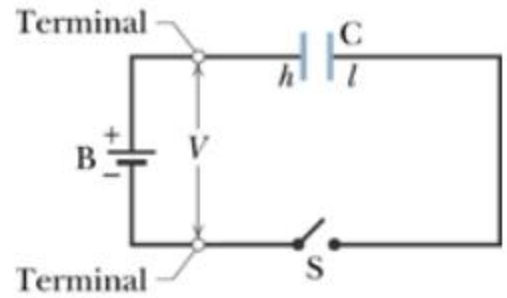


(a)

where C is the capacitance

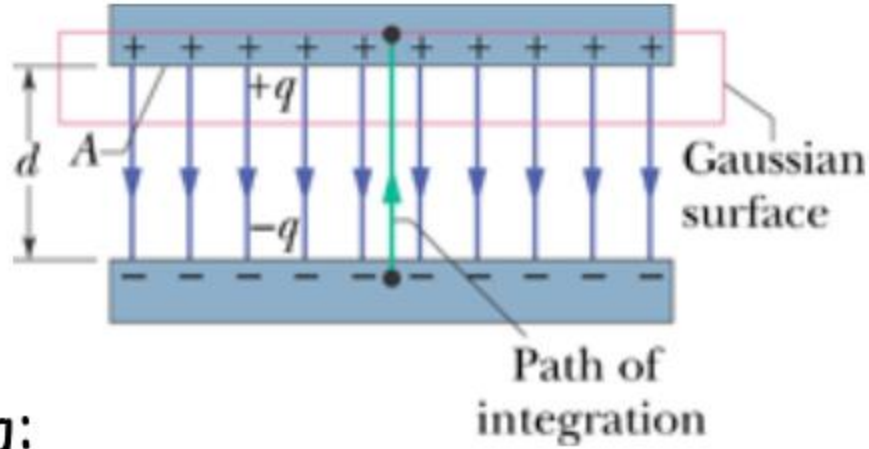
SI Unit of Capacitance: coulomb/volt= farad (F)

1 F = 10^3 mF = 10^6 μF = 10^{12} pF



(b)

THE CAPACITANCE OF A PARALLEL PLATE CAPACITOR



(1) Calculate q:

$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = q. \qquad q = \varepsilon_0 EA,$$

(2) Calculate V:

$$V = \int_{-}^{+} E\,ds = E\int_{0}^{d} ds = Ed\ .$$

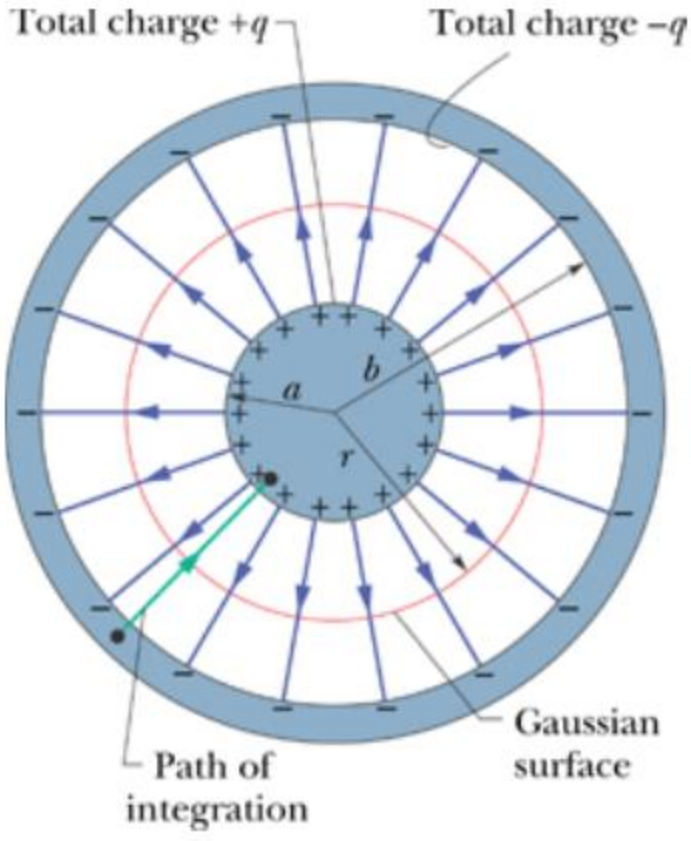
(3) Calculate C:

$$C = \frac{q}{V} = \frac{\varepsilon_0 EA}{Ed}$$

$$C = \frac{\varepsilon_0 A}{d} \text{ (parallel–plate capacitor)}\ .$$

- Only the geometry of the plates (*A* and *d*) affect the capacitance.

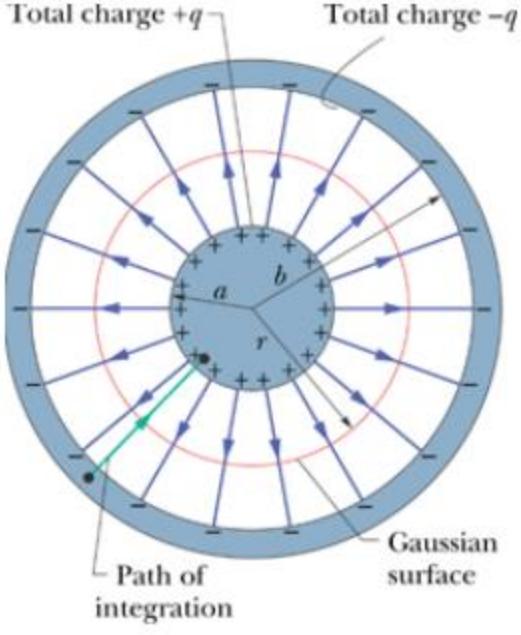
THE CAPACITANCE OF A Cylindrical Capacitor



A cylindrical capacitor of length *L* formed by two coaxial cylinders of radii *a* and *b*

$$C = 2\pi\varepsilon_0 \frac{L}{\ln(b/a)} \quad \text{(cylindrical capacitor)} .$$

THE CAPACITANCE OF A Spherical Capacitor



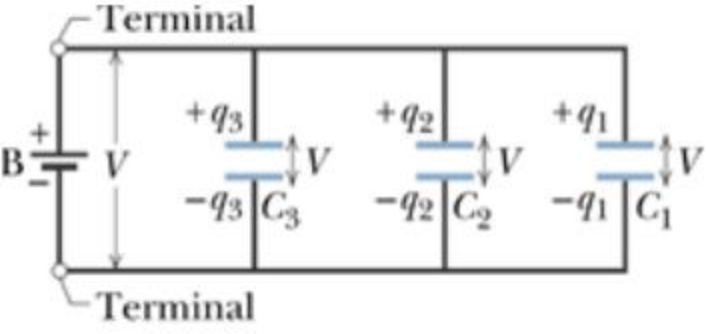
A capacitor that consists of two concentric spherical shells, of radii *a* and *b*.

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a} \text{ (spherical capacitor)}.$$

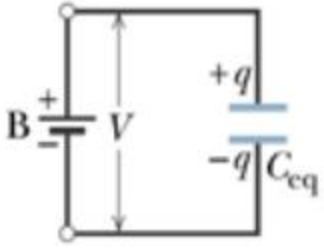
For An Isolated Sphere, a=R and b=∞

$$C = 4\pi\varepsilon_0 R \text{ (isolated sphere)}.$$

Capacitors in Parallel



(a)

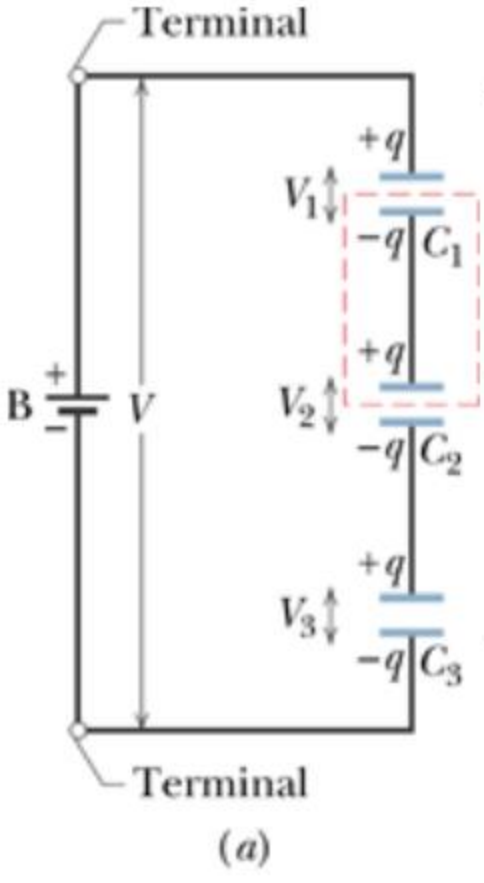


(b)

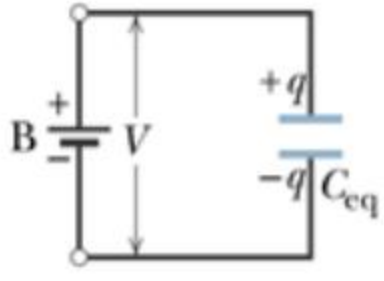
- When a potential difference *V* is applied across several capacitors connected in parallel, that potential difference *V* is applied across each capacitor.
- The total charge *q* stored on the capacitors is the sum of the charges stored on all the capacitors.
- Capacitors connected in parallel can be replaced with an equivalent capacitor that has the same *total* charge *q* and the same potential difference *V* as the actual capacitors.

$$C_{\text{eq}} = \sum_{j=1}^{n} C_j \quad (n \text{ capacitors in parallel}) .$$

Capacitors in Series



(a)



(b)

- When a potential difference V is applied across several capacitors connected in series, the capacitors have identical charge q.
- The sum of the potential differences across all the capacitors is equal to the applied potential difference V.
- Capacitors that are connected in series can be replaced with an equivalent capacitor that has the same charge q and the same *total* potential difference V as the actual series capacitors.

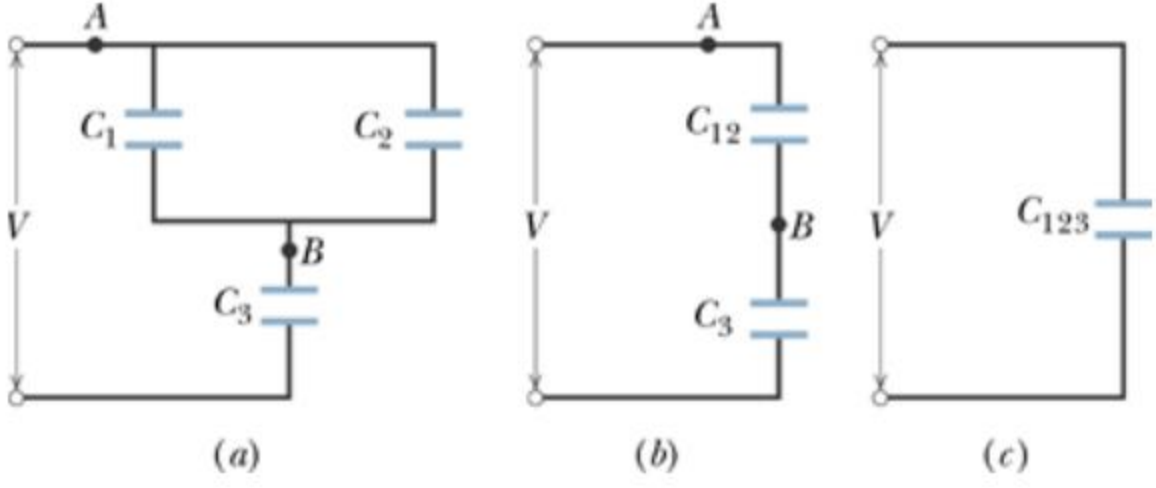
$$\frac{1}{C_{eq}} = \sum_{j=1}^{n} \frac{1}{C_j} \quad (n \text{ capacitors in series}) .$$

Sample Problem 1

(a) Find the equivalent capacitance for the combination of capacitances shown in Fig. 25-10 *a*, across which potential difference *V* is applied. Assume

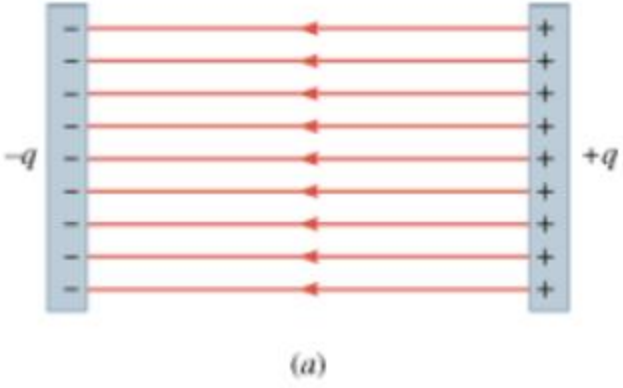
$$C_1 = 12.0\,\mu\text{F},\ C_2 = 5.30\,\mu\text{F},\ \text{and}\ C_3 = 4.50\,\mu\text{F}\ .$$

(b) The potential difference applied to the input terminals in Fig. 25-10 *a* is *V* = 12.5 V. What is the charge on *C*1?



Energy Stored in an Electric Field

The potential energy of a charged capacitor may be viewed as being stored in the electric field between its plates.



Suppose that, at a given instant, a charge q' has been transferred from one plate of a capacitor to the other. The potential difference V' between the plates at that instant will be q'/C. If an extra increment of charge dq' is then transferred, the increment of work required will be,

$$dW = V'\,dq' = \frac{q'}{C}dq' .$$

The work required to bring the total capacitor charge up to a final value q is

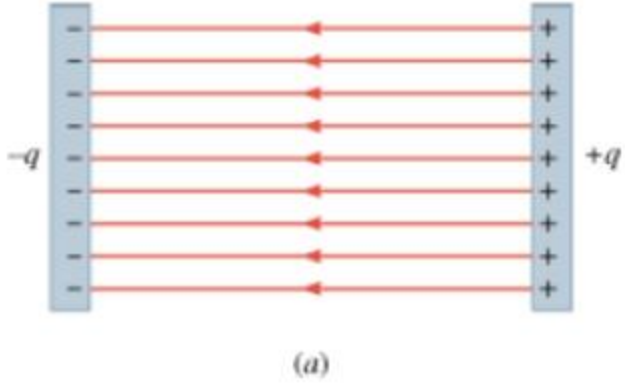
$$W = \int dW = \frac{1}{C}\int_0^q q'\,dq' = \frac{q^2}{2C} .$$

This work is stored as potential energy U in the capacitor, so that

$$U = \frac{q^2}{2C} \text{ (potential energy)} .$$ or $$U = \frac{1}{2}CV^2 \text{ (potential energy)} .$$

Energy Density

The potential energy per unit volume between parallel-plate capacitor is



(a)

$$u = \frac{U}{Ad} = \frac{CV^2}{2Ad} . \qquad (C = \varepsilon_0 A / d)$$

$$u = \frac{1}{2}\varepsilon_0 \left(\frac{V}{d}\right)^2 .$$

V/d equals the electric field magnitude *E due to* $(E = -\Delta V / \Delta s)$

$$u = \frac{1}{2}\varepsilon_0 E^2 \text{ (energy density)} .$$

Sample Problem 2

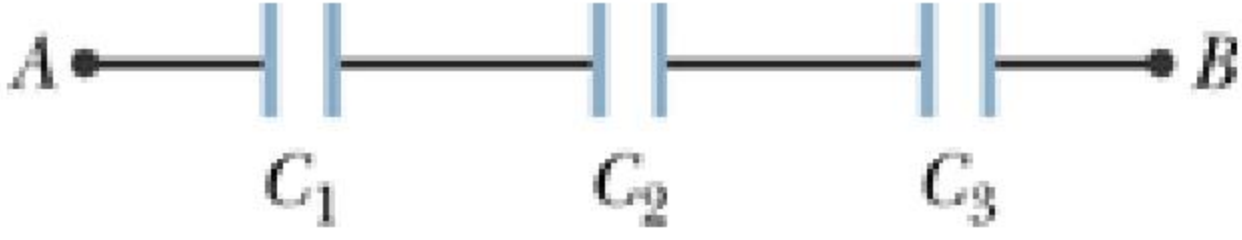
An isolated conducting sphere whose radius R is 6.85 cm has a charge q = 1.25 nC.

(a) How much potential energy is stored in the electric field of this charged conductor?

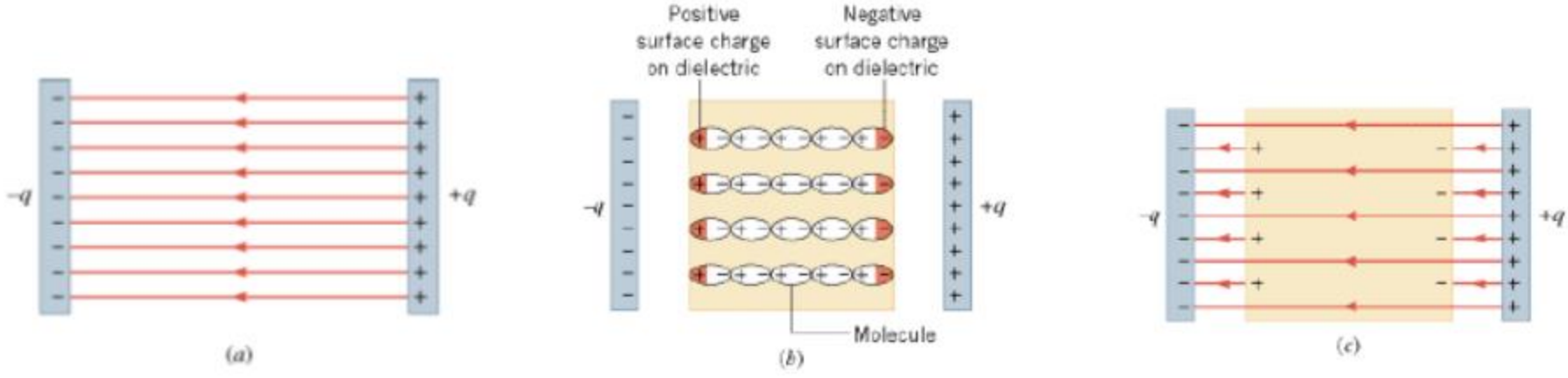
(b) What is the energy density at the surface of the sphere?

Sample Problem 3

In Fig. 25-45 , $C1$ = 10.0 μF, $C2$ = 20.0 μF, and $C3$ = 25.0 μF. If no capacitor can withstand a potential difference of more than 100 V without failure, what are (a) the magnitude of the maximum potential difference that can exist between points *A* and *B* and (b) the maximum energy that can be stored in the three-capacitor arrangement?



Capacitor with a Dielectric



THE DIELECTRIC CONSTANT

The surface charges on the dielectric reduce the electric field inside the dielectric. This reduction in the electric field is described by the ***dielectric constant*** *k*, which is the ratio of the field magnitude E_0 without the dielectric to the field magnitude E inside the dielectric:

$$\kappa = \frac{E_0}{E}$$

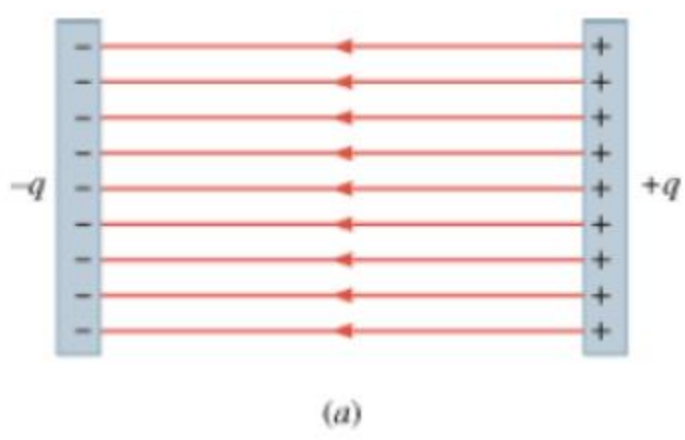
Every dielectric material has a characteristic *dielectric strength*, which is the maximum value of the electric field that it can tolerate without breakdown

Some Properties of Dielectrics

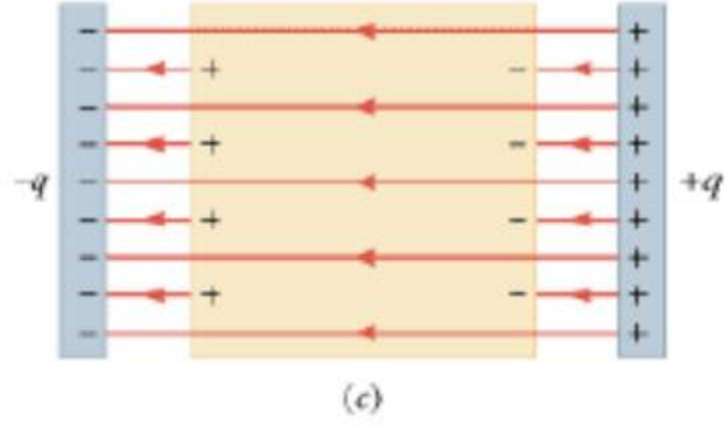
Material	Dielectric Constant	Dielectric Strength (kV/mm)
Air (1 atm)	1.00054	3
Polystyrene	2.6	24
Paper	3.5	16
Transformer oil	4.5	
Pyrex	4.7	14
Ruby mica	5.4	
Porcelain	6.5	
Silicon	12	
Germanium	16	
Ethanol	25	
Water (20°C)	80.4	
Water (25°C)	78.5	
Titania ceramic	130	
Strontium titanate	310	8

For a vacuum, $K = \text{unity}$

Capacitance with a Dielectric



(a)



(c)

$$C_{air} = \frac{q}{V} = \frac{q}{E_0 d} \qquad \kappa = E_0 / E$$

$$C_{air} = \frac{q}{E_0 d} = \frac{q}{\kappa E d} = \frac{1}{\kappa}\left(\frac{q}{Ed}\right) = \frac{C}{\kappa} \qquad C = \kappa C_{air}$$

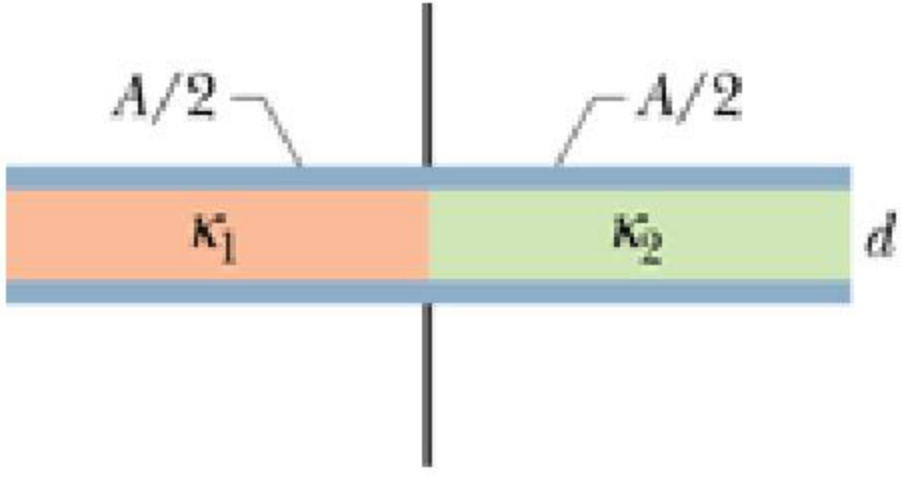
The capacitance with the dielectric present is increased by a factor of *k* over the capacitance without the dielectric.

Example 4

An empty parallel plate capacitor (C_0 = 25 mF) is charged with a 12 V battery. The battery is disconnected and the region between the plates of the capacitor is filled with pure water. What are the capacitance, charge, and voltage for the water-filled capacitor?

Example 5

Figure 25-48 shows a parallel-plate capacitor with a plate area $A = 5.56\ cm^2$ and separation $d = 5.56\ mm$. The left half of the gap is filled with material of dielectric constant $\kappa 1 = 7.00$; the right half is filled with material of dielectric constant $\kappa 2 = 12.0$. What is the capacitance?



Example 6

Figure 25-49 shows a parallel-plate capacitor with a plate area $A = 7.89\ cm^2$ and plate separation $d = 4.62$ mm. The top half of the gap is filled with material of dielectric constant $\kappa_1 = 11.0$; the bottom half is filled with material of dielectric constant $\kappa_2 = 12.0$. What is the capacitance?

